



[5. Segmentation, From Intensities To](#)
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5.2.4. Segmentation - Practice Assessment

Thresholding

0 points possible (ungraded)

Thresholding is:

☒ a simple segmentation technique ✓

☐ always the method of choice

☒ sensitive to noise ✓



Automatic Thresholding is:

☒ robust against histogram shifting ✓

☐ always the method of choice

☐ always working even when the cell density varies a lot between field of view



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i Answers are displayed within the problem

Finding a Threshold

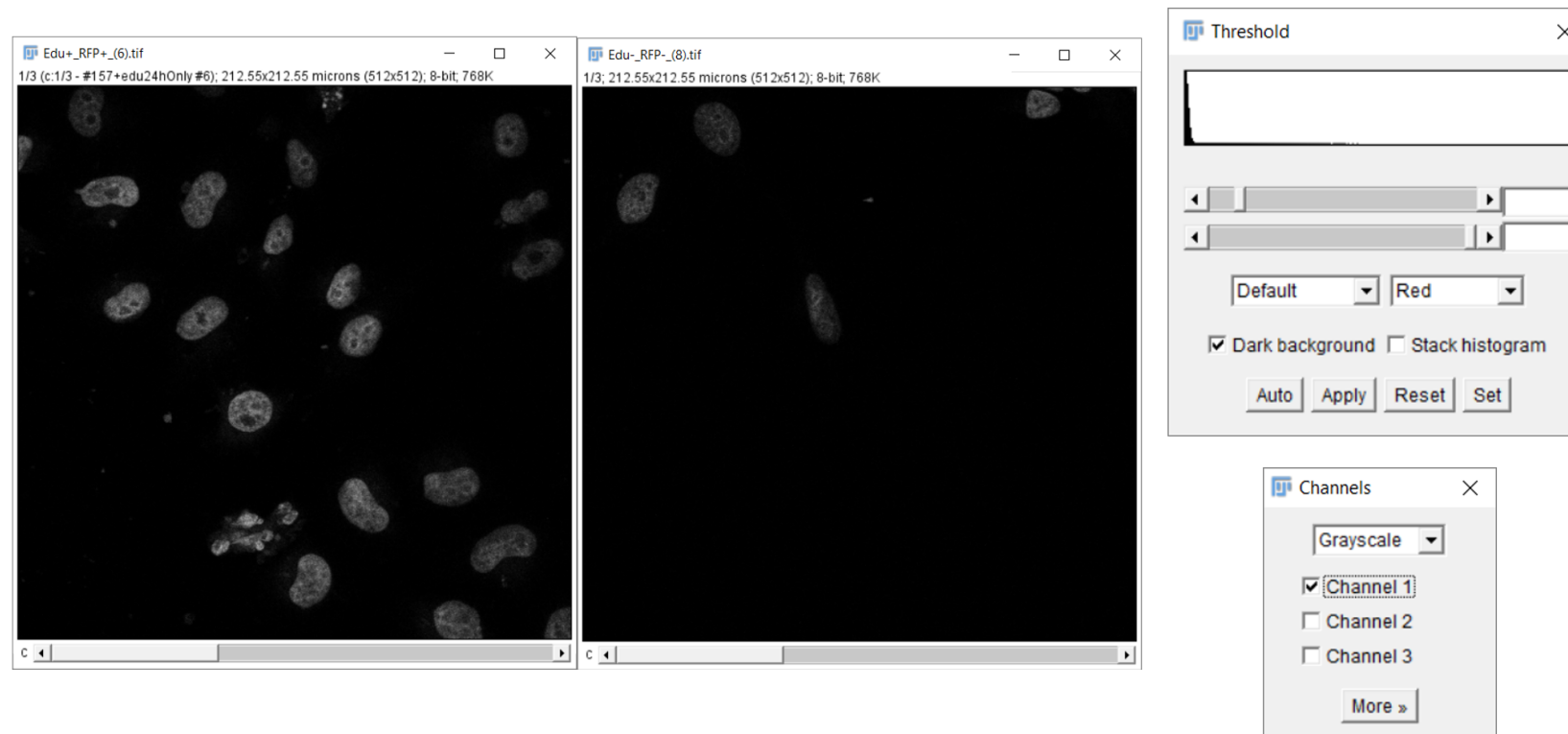


0 points possible (ungraded)

If needed, [download the HeLa cells images](#).

Open the images : "Edu+_RFP+_(6).tif" and "Edu+_RFP+_(8).tif"

Use the composite mode and select the first channel.



Which Automatic Threshold method(s) look suited to detect nuclei on **BOTH** images.

☐ Huang

☐ Percentile

☒ Intermodes ✓

☒ RenyiEntropy ✓

☐ Shanbhag



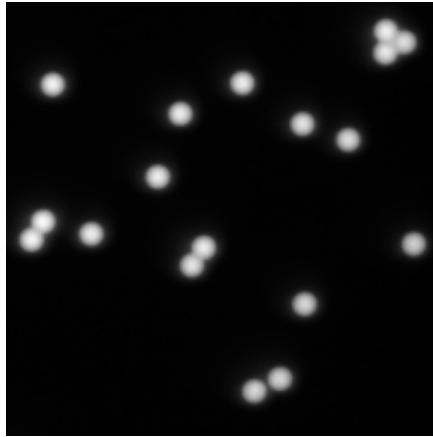
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Counting Beads

0 points possible (ungraded)

Please, download the beads images and open the image "homogeneous_beads.tif"



In the image "homogeneous_beads.tif", how many beads can you visually count ?

✓ Answer: 18

To detect and count the individual beads, a satisfying method is :

☒ Process > Find Maxima > Noise tolerance=10 ✓

☐ k-means , with 2 clusters

☐ Macro 'DifferenceOfGaussian.ijm' , with sigma1 = 1 (download the DifferenceOfGaussian.ijm)

☐ Automatic thresholding using "Default"

☐ None of the mentionned above



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